

3 Method

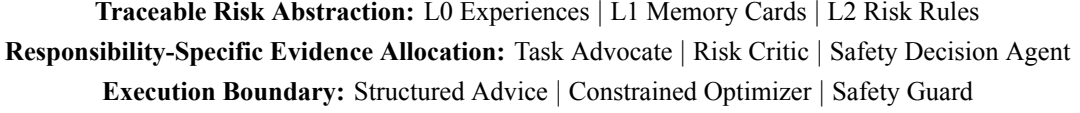


Fig. 1: Method overview. The framework constructs traceable risk evidence, allocates it by decision responsibility, and separates semantic advice from action execution.

3.1 System Overview

We propose a risk-experience-driven multi-role framework built around two coupled operations: *traceable risk abstraction* and *responsibility-specific evidence allocation*. Given interaction history $\mathcal{H}_t$, the first operation derives case-level memory cards and cross-case risk rules while retaining links to the original interaction records. Given the current planning input $\mathcal{X}_t = (g_t, r_t, e_t, \mathcal{A}_t)$, the second operation constructs a distinct evidence bundle for each decision role. Their coupling allows a decision to use compact long-term evidence without losing its empirical provenance.

The Task Advocate, Risk Critic, and Safety Decision Agent interpret their assigned evidence and return structured recommendations with cited evidence IDs. They do not issue executable controls. A constrained optimizer computes the final semantic action from the admissible set $\mathcal{A}_t$, and an independent Safety Guard validates it before execution. Section 3.2 presents the construction of traceable evidence; Section 3.3 presents its role-conditioned allocation and use.

3.2 Traceable Hierarchical Risk Memory

We construct risk evidence through successive abstraction. L0 experiences $\mathcal{E}_t$ preserve complete interactions, L1 cards $\mathcal{M}_t$ extract the risk structure of individual cases, and L2 rules $\mathcal{Q}_t$ consolidate patterns supported across cases. Stable IDs make every rule traceable through its source cards to the original records. The hierarchy therefore reduces the context required at decision time while retaining the evidence needed to inspect or revise an abstraction.

3.2.1 L0: Embodied Risk Experiences

L0 represents an interaction as E_i , which records its task and scene conditions, action A_i , outcome O_i , failure cause, risk type, severity $\sigma_i \in \{1, \dots, 5\}$, and trajectory provenance. An immutable `experience_id` links the record to sensor logs, controls, or human labels.

To compare heterogeneous observations on a common risk-oriented scale, we encode E_i as $z_i = [1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i]$. Here $\bar{c}_i$ and $\bar{d}_i$ are normalized clearance and obstacle distance, p_i is observation confidence, b_i is battery ratio, and h_i contains discrete robot, obstacle, and task conditions. The complements make larger values indicate stronger risk factors; L0 retains the full trajectory for audit.

3.2.2 L1: Risk Memory Cards

L1 compresses each experience into a risk memory card. The normalized severity $\sigma_i/5$ represents consequence magnitude, the outcome weight $\omega(O_i)$ assigns 1, 0.8, or 0.05 to failure, aborted, or success, and $\gamma_i \in [0, 1]$ discounts evidence with low record quality. Their product $q_i = (\sigma_i/5)\omega(O_i)\gamma_i$ is a relative evidence score, not a collision probability. Let F_i denote

the failure cause, B_i the mitigation and stopping conditions, and P_i the source record containing the memory ID, experience ID, and source span.

Substituting z_i and q_i into the compression map derives the L1 card:

$$\begin{aligned} M_i &= C(E_i) = \text{Pack}(z_i, A_i, O_i, F_i, B_i, q_i, P_i) \\ &= \text{Pack}([1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i], \\ &\quad A_i, O_i, F_i, B_i, \frac{\sigma_i}{5}\omega(O_i)\gamma_i, P_i). \end{aligned} \quad (1)$$

Equation (1) shows how the risk state, outcome evidence, and provenance of an L0 experience jointly form its traceable L1 card.

3.2.3 L2: Risk Rules and Cognitive Evolution

L2 groups cards with similar z_i , risk type, and action semantics into pattern set $\mathcal{I}_j$. The aggregation map $\mathcal{A}$ derives rule Q_j from their triggers, recommended and prohibited actions, mitigations, and source IDs. Let n_j^+ , n_j^- , and n_j^0 count mitigation successes, risk failures, and inconclusive aborts. We define θ_j as the probability that a new observation provides decisive evidence for rule Q_j . With a Beta(α, β) prior, its posterior mean c_j is

$$\begin{aligned} (Q_j, c_j) &= \mathcal{A}(\{M_i\}_{i \in \mathcal{I}_j}), \\ c_j &= \mathbb{E}[\theta_j \mid \mathcal{I}_j] = \frac{n_j^+ + n_j^- + \alpha}{n_j^+ + n_j^- + n_j^0 + \alpha + \beta}. \end{aligned} \quad (2)$$

Here c_j measures decisive empirical support, not hazard probability. A rule becomes *active* when its provenance count, cross-scene consistency, and reliability meet preset thresholds; conflicting or stale evidence revises its trigger or marks it *retired*. Versioned source IDs preserve the chain $Q_j \rightarrow M_i \rightarrow E_i$ through every update.

At runtime, the rule gate admits actions recommended by every matched *active* rule and prohibited by none. An empty intersection stops execution and requests human review.

3.3 Role-Conditioned Risk Evidence Retrieval

The same risk history serves different decision responsibilities. We therefore allocate evidence by role instead of supplying one shared retrieval result. The Advocate receives evidence about feasible progress and successful mitigation, the Critic receives failures and stopping conditions, and the Decision agent receives evidence needed to balance task utility and risk. Given $\mathcal{X}_t$, each card yields $\phi_i = [\text{sim}, \text{sev}, \text{rel}, \text{rec}, \text{fit}] \in [0, 1]^5$, describing contextual similarity, severity, reliability, recency, and role fit. Role-specific weights $w^{(r)}$ turn these common card features into responsibility-specific relevance.

For role r , $\mathcal{B}$ denotes a candidate evidence bundle, K limits its card count, and B_r is its token budget. The redundancy measure $d(i, j)$ discourages repeated risk types, outcomes, and

mitigations; $\hat{\tau}(M_i)$ estimates the token cost of card M_i , and λ controls the redundancy penalty. Retrieval then selects

$$\begin{aligned} \mathcal{B}_r^* = \arg \max_{\mathcal{B} \subseteq \mathcal{M}_t} & \sum_{M_i \in \mathcal{B}} (w^{(r)})^\top \phi_i - \lambda \sum_{\substack{M_i, M_j \in \mathcal{B} \\ i < j}} d(i, j) \\ \text{s.t. } |\mathcal{B}| \leq K, & \sum_{M_i \in \mathcal{B}} \hat{\tau}(M_i) \leq B_r. \end{aligned} \quad (3)$$

We use $w^{(\text{adv})} = (.35, .05, .20, .05, .35)$, $w^{(\text{crit})} = (.30, .25, .15, .05, .25)$, and $w^{(\text{dec})} = (.35, .20, .20, .10, .15)$ for (sim, sev, rel, rec, fit). Algorithm 1 greedily approximates Eq. (3); the experiments evaluate the role-specific weighting.

Each bundle stores its role and retrieval version, selected L1 cards and L2 rules, scores, rationales, budget use, and source IDs. Agents declare used evidence through `cited_experience_ids` and `cited_rule_ids`; the framework rejects IDs absent from the bundle, preserving a complete retrieval-to-decision audit trail.

Let $\mathcal{D}_t$ denote the three role reports and $\mathcal{F}_t$ the actions that satisfy hard constraints and matched L2 prohibitions. For action a , $U(a)$ is task utility, $R(a \mid \mathcal{B}_{\text{dec}}^*)$ is the combined scene-memory risk, and $Z(a \mid \mathcal{D}_t, \mathcal{B}_{\text{dec}}^*)$ is uncertainty from observation confidence, memory coverage, and role disagreement. The thresholds ρ and η bound these relative scores. The deter-

ministic optimizer selects

$$\begin{aligned} a_t^* &= \arg \max_{a \in \mathcal{F}_t} U(a), \\ R(a \mid \mathcal{B}_{\text{dec}}^*) &\leq \rho, \\ Z(a \mid \mathcal{D}_t, \mathcal{B}_{\text{dec}}^*) &\leq \eta. \end{aligned} \quad (4)$$

Rule conflict, invalid output, an empty feasible set, or Guard rejection triggers `safe_stop`, `observe_again`, or human review. This execution boundary lets the agents interpret retrieved evidence while deterministic modules retain action authority.

Algorithm 1: Role-Conditioned Risk Retrieval

Require: State $\mathcal{X}_t$, roles $\mathcal{R}$, L1 cards $\mathcal{M}$, active L2 rules $\mathcal{Q}$, budgets $\{B_r\}$, limit K
Ensure: Role-specific evidence bundles $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$

- 1: Validate $\mathcal{X}_t$, $\mathcal{R}$, $\mathcal{M}$, and $\mathcal{Q}$
- 2: **for** each role $r \in \mathcal{R}$ **do**
- 3: Compute all ϕ_i ; initialize candidates $\mathcal{C}_r$ and $\mathcal{B}_r \leftarrow \emptyset$
- 4: **while** $\mathcal{C}_r \neq \emptyset$ and $|\mathcal{B}_r| < K$ **do**
- 5: Select M_{i^*} by marginal gain in (3)
- 6: Remove M_{i^*} from $\mathcal{C}_r$
- 7: **if** $\hat{\tau}(\mathcal{B}_r) + \hat{\tau}(M_{i^*}) \leq B_r$ **then**
- 8: $\mathcal{B}_r \leftarrow \mathcal{B}_r \cup \{M_{i^*}\}$
- 9: **end if**
- 10: **end while**
- 11: Append matched active L2 rules, scores, rationales, and source IDs to $\mathcal{B}_r$
- 12: **end for**
- 13: **return** $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$